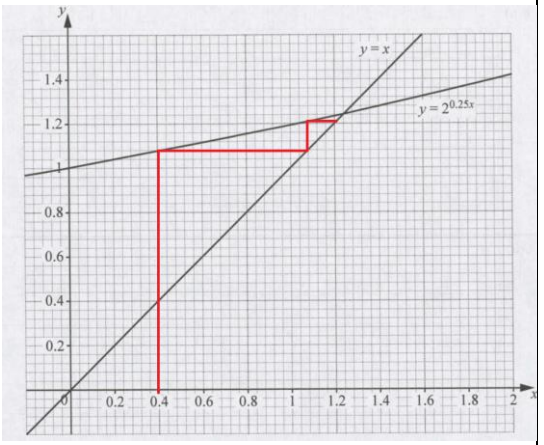
Question			Answer	Marks	AO	Guidance
5	(a)		**DR** $1^4 - 2^1 \quad (= -1 < 0)$ and $2^4 - 2^2 \quad (= 12 > 0)$	**M1**	**1.1**	**See appendix** • Allow $1^4 - 2^1$ or 1 – 2 **and** $2^4 - 2^2$ or 16 – 4 • Ignore '= 0' • Condone -1 and 12 only provided clearly linked to $x = 1$ and $x = 2$ e.g. as coordinates or in a table of values • Allow for using two values between 1 and 2 that are either side of 1.24 e.g. 1.2 and 1.3
			Values are -1 and 12 so there is a change of sign (and hence a root between 1 and 2)	**A1** **[2]**	**2.4**	• Correct values of function **and** comment on change of sign, allow e.g. $-1 < 0$ and $12 > 0$ • Condone no mention of function being sufficiently well-behaved (or continuous) • e.g. 1.2 and 1.3 leading to -0.22 and 0.39 (2 s.f. is sufficient)
5	(b)		$x^4 = 2^x$ $x = (2^x)^{\frac{1}{4}}$ $= 2^{0.25x}$	**B1** **[1]**	**2.1**	• AG • **Convincing** rearrangement to given result • Must show one step of working and no incorrect work seen • Must not work backwards • e.g. $4\log_2(x) - x = 0$ is **B0** as not convincingly from a correct process unless $4\log_2(x) = x$ is seen first
5	(c)		x_1 =1.07…. 1.24	**M1** **A1** **[2]**	**1.1** **2.2a**	• x_1 = awrt 1.07 (= 1.07177...). • soi by any of these values: 1.204…,1.232…,1.237… • cao • Must have shown at least one iteration e.g. x_1 =1.07

Question			Answer	Marks	AO	Guidance
5	(d)			**B1** **B1** **[2]**	**1.1** **1.1**	• Vertical line $x = 0.4$ drawn to meet the curve, could start line from $y = x$ • Correct completion of staircase diagram to head towards root with at least two vertical and two horizontal lines seen (lines need not extend beyond region between line and curve) • Condone good freehand lines, mark intent • **SC B1B0** for using a different starting value e.g. $x = 0$ with at least two vertical and two horizontal lines seen. Correct staircase diagram converging to root from *their* starting value.
5	(e)		$x^4 = 2^x$ $x = \log_2(x^4)$ $= 4\log_2 x$	**B1** **[1]**	**2.1**	• AG • **Convincing** rearrangement to given result • Must show at least one step of working and no incorrect work seen • e.g. $\log_2(x^4) - \log_2(2^x) = 0$ is **B0** as not convincingly from a correct process unless $\log_2(x^4) = \log_2(2^x)$ is seen first • Must not work backwards
5	(f)		**DR** $x_1 = 0$ Not possible to find log of zero	**M1** **A1** **[2]**	**1.1** **2.4**	• Or $4\log_2 1 = 0$ • Condone e.g. math error or x_2 is invalid or x_2 has no solution provided log 0 is mentioned

Appendix

Question 5a

REMEMBER: It is not possible to have an award of M0A1

After M1 has been awarded the following comments would earn A1
There is a sign change therefore the curve crosses the x-axis therefore a root is between 1 and 2
Change of sign and root must be between 1 and 2
$-1 < 0$ and $12 > 0$
-1 is negative and 12 is positive
Change of sign then there is a root between 1.2 and 1.3, which also lies between 1 and 2

After M1 has been awarded the following comments would earn A0
$x = 1, y = -1$ and $x = 2, y = 8$, sign change suggests a root between these values **A0 as $y = 8$ is incorrect**
There is a root between 1 and 2 **A0 as no mention of sign change**